

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER

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ARTIFACT

Problem Brief

STATUS

Working document

SECTION 01

Visible Failure

MR-17 has no active test assignment assigned to it. However, at about 2:13 a.m., MR-17 began repeating Test K-44, a test that does not appear in the current test catalog. It began its repeat cycle at 02:13, and returned an "Active Instruction Pending" status when an operator stop request was issued at 02:14. The system received a validly formatted command "legacy_sync.exe" through an accepted internal interface, but telemetry does not identify the service owner. Staff statements note that MR-17 behaved normally beforehand, production has paused in the affected QA cell, and a controlled restart might clear the active state.

SECTION 02

Verified Facts

- 1.MR-17 accepted Test K44 at 02:13 as recorded in the QA Cell Incident Log by Nia Voss.
- 2.Test K-44 was marked retired 186 days before the incident according to the maintenance archive records terminal.
- 3.The command received by the robot was validly formatted and came from "legacy_sync.exe" as written in the command telemetry by Elias Chen.

SECTION 03

Assumptions and Open Questions

Assumption: legacy_sync.exe caused the entire incident.

Question: Was Test K-44 removed from every service and schedule

SECTION 04

System Environment Map

SECTION 04A

People and Roles

- 1.Nia Voss: Night QA Supervisor, who recorded the initial incident log entries regarding MR-17 and reported that the robot behaved normally before the incident.
- 2.Elias Chen: Robotics Systems Engineer, who manages or analyzes command telemetry and noted that a controlled restart may clear the active state.
- 3.Mara Ilyan: Production Operations Manager, who reported that production has paused in the affected QA cell.
- 4.Operators: Personnel who issue stop requests and interact with the QA cell systems (e.g the operator who attempted to stop MR-17).

SECTION 04B

Processes and Rules

- 1.The active scheduling console manages test assignments and catalogs.
- 2.The maintenance archive tracks retired tests and workflow removals.
- 3.Internal command interfaces process validly formatted commands like `legacy_sync.exe`.
- 4.QA cell operating procedures allow operators to issue stop requests.

SECTION 04C

Information and Data

- 1.The QA Cell Incident Log matters because it tracks the timeline of MR-17 and the failed stop request.
- 2.The Current Scheduler records terminal matters because it shows no active test assignment for MR-17.
- 3.The Maintenance Record matters because it shows Test K-44 was retired 186 days before the incident.
- 4.The Command Telemetry matters because it shows a valid command came from legacy_sync.exe.

SECTION 04D

Technology and Tools

- 1.The current scheduling console manages active test assignments.
- 2.The engineers process command telemetry.
- 3.Internal command interfaces handle validly formatted commands.
- 4.The legacy_sync.exe service interacts with the system.

SECTION 04E

Environment and Constraints

- 1.Production has paused in the affected QA cell.
- 2.The operator stop request returned active instruction pending.
- 3.The current telemetry does not identify the service owner.

SECTION 05

Investigation Boundary

Inside the Investigation

Based on the provided case materials, the team will examine the unverified scope to determine if every copy, service, or schedule of Test K-44 was removed.

Outside for Now

Current evidence does not require recommending a repair yet because the case materials instruct that analysts should not recommend a repair in Week 1.

SECTION 06

Organizational Impact

1. Production is paused in the affected QA cell.
2. MR-17 is affected by an unstopped repeat cycle after accepting "TEST_K44_START"

SECTION 07

Problem Statement

The gap is that it remains unverified whether every copy, service, or schedule of Test K-44 was completely removed after being marked retired 186 days prior. The impact of this gap is that production has paused in the affected QA cell, and the operator's stop request returned an active instruction pending status.

SECTION 08

Next Investigation Step

Action: Examine the unverified scope across active nodes and registries.

Source: Maintenance archive and the Current Scheduler.

Question: Was every copy, service, or schedule of Test K-44 fully removed from the system?

SHARED ARTIFACT

Working Document Link

https://github.com/nanyanwu1985/CIT140-Portfolio/blob/main/Week01_Something_Isnt_Adding_Up/artifact/CIT-140%20Week%201%20Problem%20Brief.pdf

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.